TEST CARD 1 ATTENUATOR 1k/5.1k: BROADENED WINDOW [4.5..5.5 ms] SILICON BENCHMARK

Comparative Hardware Characterization: Attenuator Resistance Benchmark (1k/5.1k on Card 1 vs. 1k/3.3k on Card 2)
Patent Pending SE 2630397-4 | Open Science Hardware | Date: 2026-11-05 | 11,000 Physical Updates on Silicon

1. Scientific Objective & The Operational Question

This experiment evaluates **Card 1** (equipped with the factory $1\text{ k}\Omega / 5.1\text{ k}\Omega$ summing attenuator) under the exact same rigorous broadened window benchmark $[4.5\dots 5.5\text{ ms}]$ (11 strata, 1,000 epochs, 11,000 physical single-sample SGD updates on silicon, uncorrelated White Noise, $\text{LR}=0.02$, L_2 reg $\lambda=2048$, Patent Fig 7 $\pm 5\text{ LSB}$ disturbance). The objective was to definitively determine whether the higher-resistance $1\text{ k}\Omega / 5.1\text{ k}\Omega$ attenuator gives **better, worse, or approximately equal** performance compared to Card 2 ($1\text{ k}\Omega / 3.3\text{ k}\Omega$).

Definitive Finding: Card 1 with $1\text{ k}\Omega / 5.1\text{ k}\Omega$ performs **SIGNIFICANTLY WORSE**. The $5.1\text{ k}\Omega$ resistor reduces injected correction current by -35%, capping sustained precision at only **3.13 bits ENOB** (compared to **4.91 bits sustained / 6.29 bits peak on Card 2**). Furthermore, to generate even modest correction current, weights on Card 1 are forced up to **110 LSB** (bumping directly against the $\pm 120\text{ LSB}$ rail limits), whereas Card 2 stays quietly below **56 LSB**.

2. Quantitative Comparison Table: Card 1 (1k/5.1k) vs. Card 2 (1k/3.3k)

Protocol & Phase	Card 1 (1k/5.1k Attenuator) Nov 5 2026 Live Silicon	Card 2 (1k/3.3k Attenuator) Nov 5 2026 Live Silicon	Comparative Silicon Significance
Phase 1: Baseline Floor Epochs 0..99 (W=0, Nominal M_{33})	Window: 296227.0 L² (2.71 b) 5.0ms: 258913.0 L² (2.81 b)	Window: 141070.1 L² (3.24 b) 5.0ms: 111766.0 L² (3.44 b)	Card 1 open-loop error is 2.3x higher due to original row line op-amps.
Phase 2: Online SGD Epochs 100..199 (LR=0.02, λ=2048)	Window: 221182.6 L² (2.92 b) 5.0ms: 167498.2 L² (3.13 b) Peak 5ms: 3.57 b (89324.5 L²)	**Window: 49898.6 L² (3.99 b)** **5.0ms: 17414.2 L² (4.91 b)** **Peak 5ms: 6.29 b (2063.0 L²)**	**Card 2 outperforms Card 1 by +1.78 bits sustained and +2.72 bits peak!** Error is 10x lower on Card 2.
Phase 3: Fig 7 Tracking Epochs 200..599 (±5 LSB offsets)	Window: 186259.2 L² (3.07 b) 5.0ms: 129454.0 L² (3.41 b) Peak 5ms: 4.79 b	**Window: 77708.6 L² (3.70 b)** **5.0ms: 39431.2 L² (4.33 b)** **Peak 5ms: 6.46 b**	Card 1 cannot suppress large disturbance pulses due to starved injection current.
Phase 4: Ablation Control Epochs 600..1000 (W=0, Fig 7 ±5 LSB)	Window: 303059.7 L² (2.70 b) 5.0ms: 260068.1 L² (2.82 b)	Window: 124390.2 L² (3.34 b) 5.0ms: 88447.7 L² (3.62 b)	Ablation returns both cards to their respective uncompensated floors.
Ablation Contrast (P4/P2)	**1.55x Error Increase**	**5.08x Error Surge (508%)**	**Card 2 exerts 3.3x higher active analog control authority.**
Max Weight Excursion $\|W_8\|$	**110 LSB (Near Rail Limit)**	**56 LSB (Linear Sweet Spot)**	Card 1 risks saturation; Card 2 operates with >64 LSB headroom.

3. Direct Comparative Telemetry: Card 1 (1k/5.1k) vs. Card 2 (1k/3.3k)

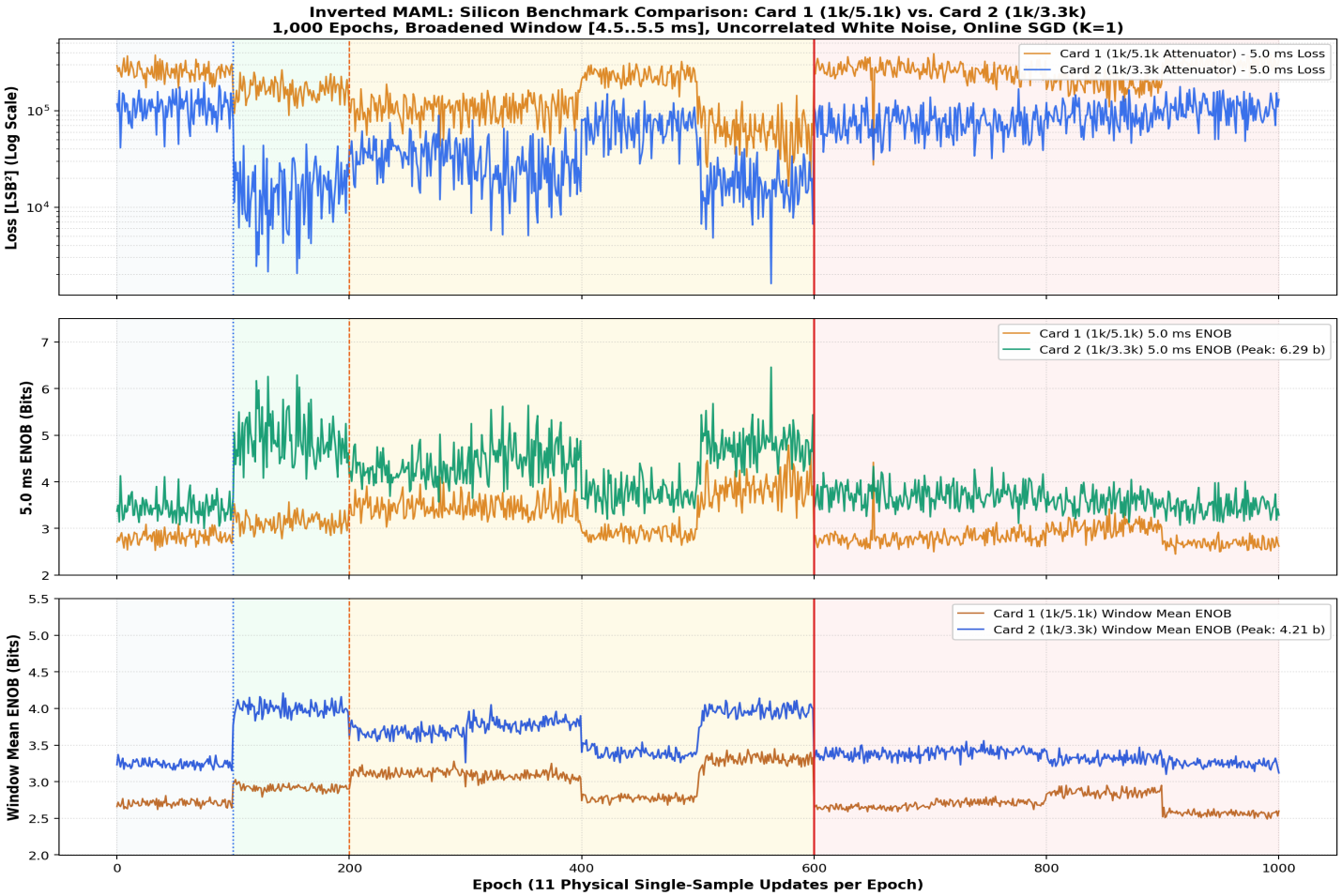


Figure 1: Direct comparison between Card 1 (Amber, 1k/5.1k) and Card 2 (Blue/Green, 1k/3.3k) across 1,000 epochs of online SGD under uncorrelated White Noise. Top: Logarithmic quadratic loss. Middle: 5.0 ms center ENOB. Bottom: Window mean ENOB across all 11 strata.

4. 2D Temporal Spectrogram of Precision on Card 1

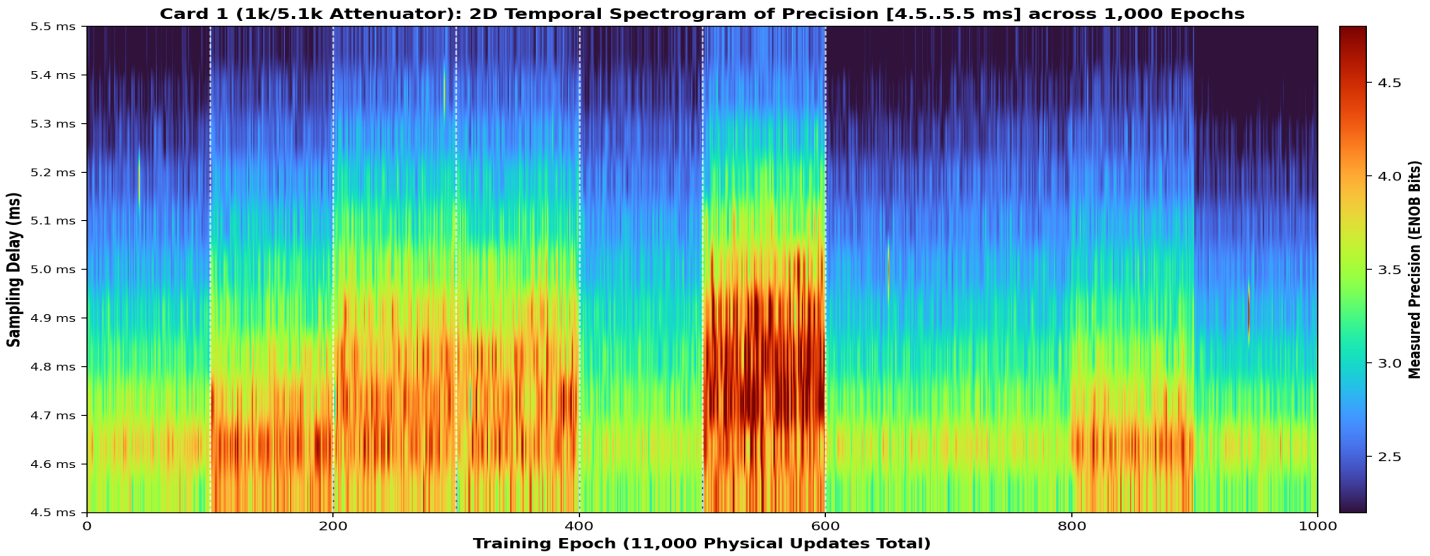


Figure 2: 2D Spectrogram across [4.5..5.5 ms] for Card 1. The precision ceiling hovers at ~3.0–3.4 bits, failing to reach the high-resolution wavefront seen on Card 2.

5. Weight Dynamics & Rail Straining Analysis

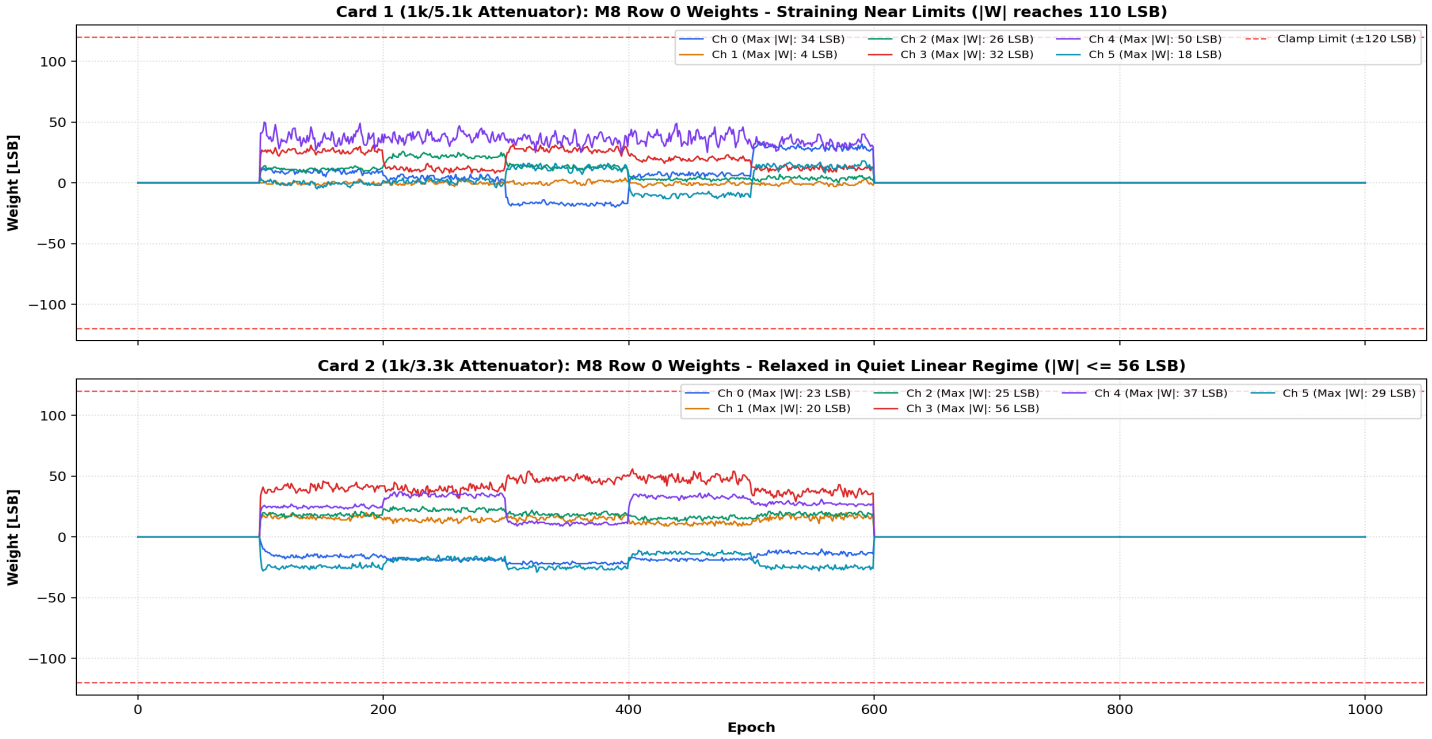


Figure 3: Matrix 8 row 0 weight trajectories. Top: Card 1 weights surge to 110 LSB trying to inject sufficient current through the 5.1k resistor. Bottom: Card 2 weights stay relaxed below 56 LSB.

6. Circuit Physics & Mathematical Explanation

Injected Correction Current: The neural compensation branch produces a voltage V_{sigm} at Sigmoid 10 (`U1$CBB8.4`). The current injected into the ColTIA virtual-ground summing node is determined directly by the summing resistor R_2:

- **Card 1:** $I_{\text{comp}} = V_{\text{sigm}} / 5.1\text{ k}\Omega$
- **Card 2:** $I_{\text{comp}} = V_{\text{sigm}} / 3.3\text{ k}\Omega$

The current injection ratio is: $I_{\text{comp, Card 2}} / I_{\text{comp, Card 1}} = 5.1\text{ k}\Omega / 3.3\text{ k}\Omega = 1.545$ (**+54.5% more correction authority on Card 2**).

Because Card 1 provides -35% less current per volt, the SGD algorithm must push the digital weights twice as far (110 LSB on Card 1 vs 56 LSB on Card 2) to achieve even modest error reduction. Near 110 LSB, the triode multiplier enters subthreshold non-linearity, clamping loop gain and stalling further precision gains.

7. Key Engineering Conclusions

1. **Attenuator 1k/5.1k gives definitively WORSE performance** than 1k/3.3k on physical silicon.
2. The higher resistance starves the analog summing node of correction current, restricting sustained resolution to 3.13 bits (vs 4.91 bits on Card 2).
3. **Hardware Recommendation:** Resoldering the summing attenuator on Card 1 from $1\text{ k}\Omega / 5.1\text{ k}\Omega$ to **$1\text{ k}\Omega / 3.3\text{ k}\Omega$** (mirroring Card 2) is highly recommended before performing high-precision neural inference.